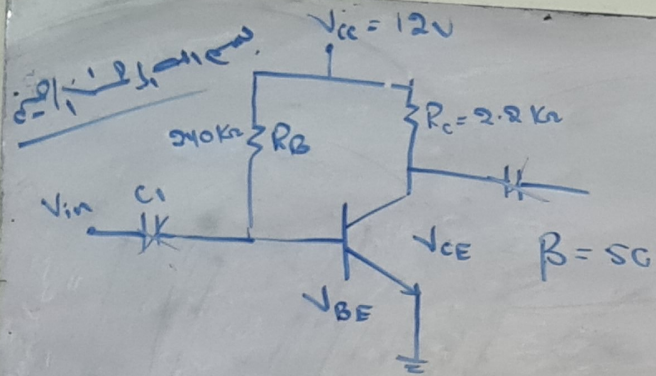




$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$= 12 - 2.355mA \times 2.2k = \underline{6.819V}$$

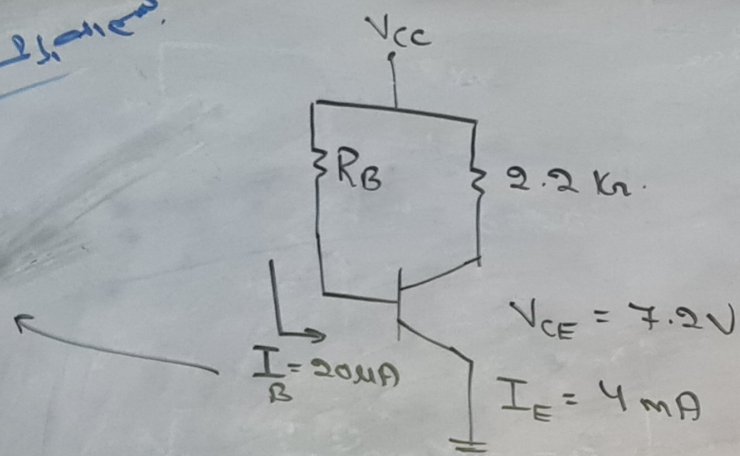
$$V_E = I_E R_E = 0$$

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 - 0.7}{240k} = \underline{0.0471mA} = 47.1\mu A$$

$$I_C = \beta I_B = 50 \times 0.0471 = \underline{2.355mA}$$

$$I_E \approx I_C \rightarrow \underline{I_E = 2.355mA}$$



Find
 ① V_{CC} , ② R_B and ③ β

$$I_C \approx I_E \rightarrow \underline{I_C = 4 mA}$$

$$[V_{CC} - I_C R_C - V_{CE} = 0]$$

$$V_{CC} = I_C R_C + V_{CE}$$

$$V_{CC} = 4 mA \times 2.2 k\Omega + 7.2 = 16 V$$

$$\therefore [I_C = \beta I_B]$$

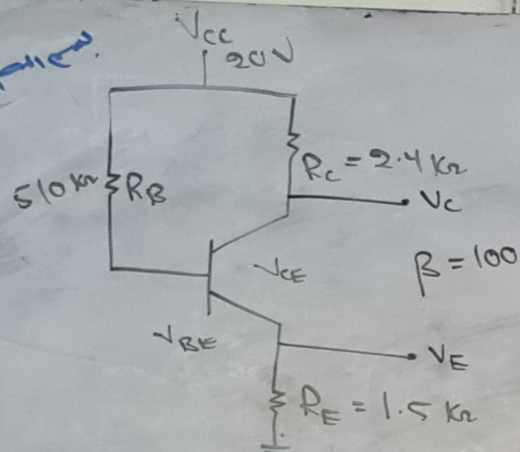
$$\beta = \frac{I_C}{I_B} = \frac{4 \times 10^{-3}}{20 \times 10^{-6}} = 200$$

$$[V_{CC} - I_B R_B - V_{BE} = 0]$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{16 - 0.7}{20 \times 10^{-6}} = 765000 \Omega$$

$\hookrightarrow 765 k\Omega$

مسألة ١٠



Find

I_B, I_C, V_{CE}, V_C and V_E

$$* V_{CC} - I_C R_C - V_C = 0$$

$$[V_C = V_{CC} - I_C R_C] = 20 - 2.918 \text{ mA} \times 2.4 \text{ k} = \underline{12.99 \text{ V}}$$

$$* V_E - I_E R_E = 0$$

$$[V_E = I_E R_E] = 2.918 \text{ mA} \times 1.5 \text{ k} = \underline{4.37 \text{ V}}$$

$$* V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E = (\beta + 1) I_B$$

$$V_{CC} - I_B R_B - V_{BE} - I_B R_E (\beta + 1) = 0$$

$$* \left[I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} \right] = \frac{20 - 0.7}{510 + 101 \times 1.5 \text{ k}} = 0.02918 \text{ mA}$$

$$I_C = \beta I_B = 100 \times 0.02918 \text{ mA} = 2.918 \text{ mA}$$

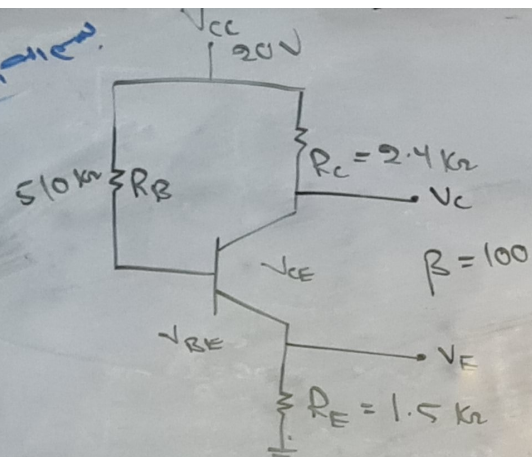
$$* V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$I_E = I_C$$

$$V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0$$

$$* [V_{CE} = V_{CC} - I_C (R_C + R_E)] = 20 - 2.918 \text{ mA} (2.4 + 1.5) \text{ k} = \underline{8.6 \text{ V}}$$

مشكلة ١٠



Find

I_B, I_C, V_{CE}, V_C and V_E

$$* V_{CC} - I_C R_C - V_C = 0$$

$$[V_C = V_{CC} - I_C R_C] = 20 - 2.918 \text{ mA} \times 2.4 \text{ k}\Omega = \underline{12.99 \text{ V}}$$

$$* V_E - I_E R_E = 0$$

$$[V_E = I_E R_E] = 2.918 \text{ mA} \times 1.5 \text{ k}\Omega = \underline{4.37 \text{ V}}$$

$$* V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E = (\beta + 1) I_B$$

$$V_{CC} - I_B R_B - V_{BE} - I_B R_E (\beta + 1) = 0$$

$$* \left[I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} \right] = \frac{20 - 0.7}{510 + 101 \times 1.5 \text{ k}} = 0.02918 \text{ mA}$$

$$I_C = \beta I_B = 100 \times 0.02918 \text{ mA} = 2.918 \text{ mA}$$

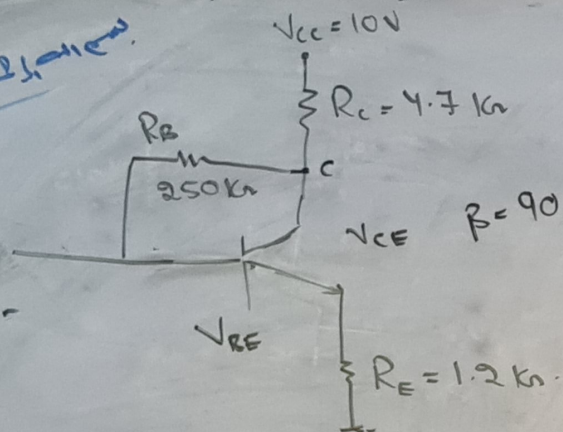
$$* V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$I_E = I_C$$

$$V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0$$

$$* [V_{CE} = V_{CC} - I_C (R_C + R_E)] = 20 - 2.918 \text{ mA} (2.4 + 1.5) \text{ k} = \underline{8.6 \text{ V}}$$

مسألة ١٠



I_C , V_{CE}

$$* V_{CC} - I_C R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E \approx I_C = \beta I_B$$

$$V_{CC} - I_B \beta R_C - I_B R_B - V_{BE} - I_B \beta R_E = 0$$

$$\left[I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} \right] = \frac{10 - 0.7}{250 + 90(4.7 + 1.2)} = 0.0119 \text{ mA}$$

$$I_C = \beta I_B = 90 \times 0.0119 = 1.071 \text{ mA}$$

$$* V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$I_E = I_C$$

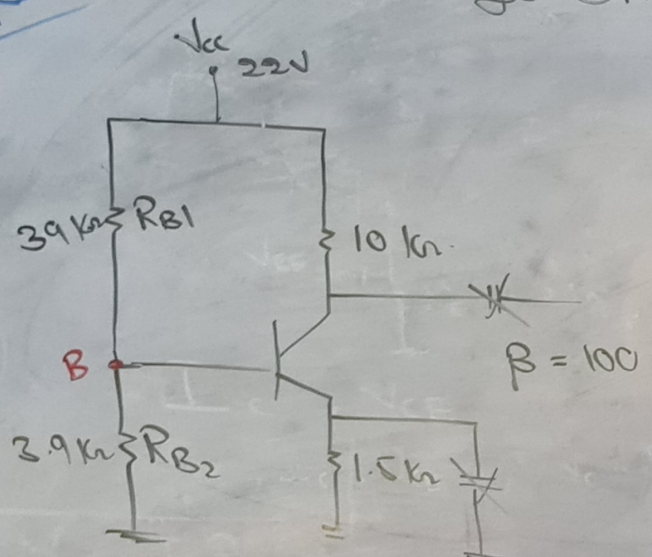
$$V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0$$

$$\left[V_{CE} = V_{CC} - I_C (R_C + R_E) \right] = 10 - 1.071 \text{ mA} \times (4.7 + 1.2) \text{ k} = 3.63 \text{ V}$$

4.0 ± V

سلسلة المقاومات

Voltage Divider $\rightarrow$ more stable



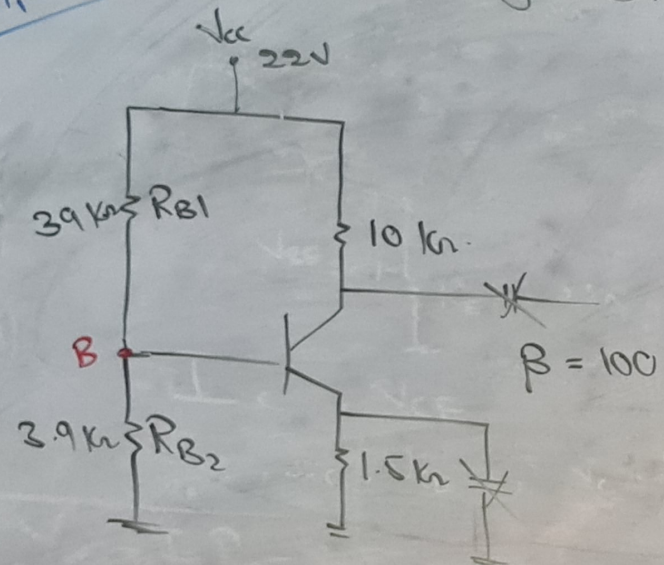
$$R_{th} = \frac{R_{B1} \times R_{B2}}{R_{B1} + R_{B2}} = \frac{39 \times 3.9}{39 + 3.9} = 3.55 \text{ k}\Omega$$

$$V_{th} = \frac{V_{cc}}{R_{B1} + R_{B2}} \times R_{B2} = \frac{22}{(39 + 3.9) \text{ k}} \times 3.9 \text{ k} = 2 \text{ V}$$

Find I_c and V_{CE}

مخطط الدارة

Voltage Divider $\rightarrow$ more stable

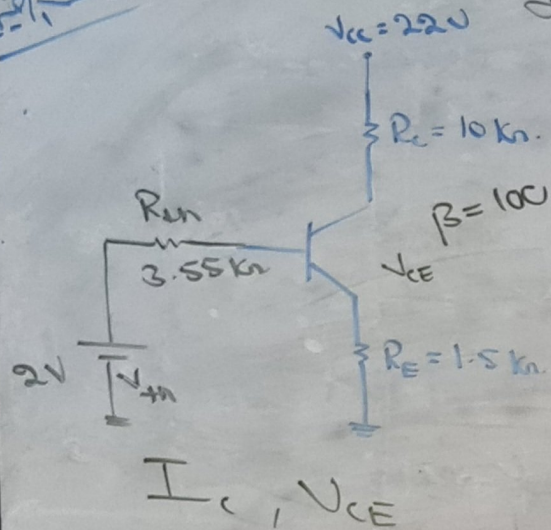


$$\left\{ \begin{aligned} R_{th} &= \frac{R_{B1} \times R_{B2}}{R_{B1} + R_{B2}} = \frac{39 \times 3.9}{39 + 3.9} = 3.55 \text{ k}\Omega \\ V_{th} &= \frac{V_{cc}}{R_{B1} + R_{B2}} \times R_{B2} = \frac{22}{(39 + 3.9) \text{ k}} \times 3.9 \text{ k} = 2 \text{ V} \end{aligned} \right.$$

Find I_c and V_{CE}

سوال ۱۱

Voltage Divider $\rightarrow$ more stable



$$V_{th} - I_B R_{th} - V_{BE} - I_E R_E = 0$$

$\downarrow$
 $I_B (\beta + 1)$

$$V_{th} - I_B R_{th} - V_{BE} - I_B (\beta + 1) R_E = 0$$

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) R_E} = \frac{2 - 0.7}{(3.55 + 101 \times 1.5)k} = 0.00838mA$$

$$I_c = \beta I_B = 100 \times 0.00838 = 0.838mA$$

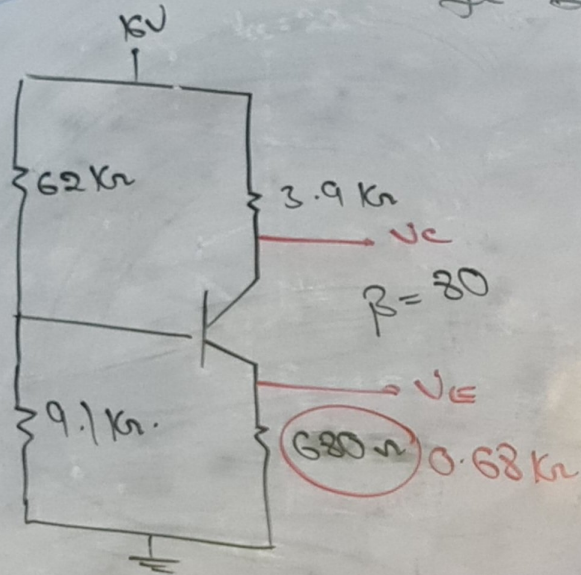
$$V_{cc} - I_c R_c - V_{ce} - I_E R_E = 0$$

$\downarrow$
 I_c

$$V_{ce} = V_{cc} - I_c (R_c + R_E) = 22 - 0.838mA \times (10 + 1.5)k = 12.363V$$

سلسلة المقاومات

Voltage Divider → more stable



$$R_{th} = \frac{62 \times 9.1}{62 + 9.1} = 7.935 \text{ k}\Omega$$

$$V_{th} = \frac{15}{(62 + 9.1) \text{ k}\Omega} \times 9.1 \text{ k}\Omega = 2.048 \text{ V}$$

$$V_c = V_{cc} - I_c R_c$$

$$V_e = I_e R_e$$

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1)R_e} =$$

$$I_c = \beta I_B =$$

$$V_{CE} = V_{cc} - I_c (R_c + R_e) =$$